

BREGENZ, Austria

WMO: 111010

Lat: 47.4992N

Lon: 9.7461E

Elev: 1391

StdP: 13.97

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 17.2	(c) 21.0	(d) 12.3	(e) 10.9	(f) 18.5	(g) 16.4	(h) 13.3	(i) 22.8	(j) 15.4	(k) 40.3	(l) 13.9	(m) 39.7	(n) 3.3	(o) 190	(p) 0.453

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 15.0	(c) 84.3	(d) 68.4	(e) 81.2	(f) 67.4	(g) 78.5	(h) 66.2	(i) 70.3	(j) 81.0	(k) 68.7	(l) 78.6	(m) 67.3	(n) 76.3	(o) 4.4	(p) 330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.6	103.0	75.4	65.2	98.1	73.7	63.9	93.5	72.1	35.1	81.4	33.7	78.8	32.5	76.6	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
14.1	12.0	10.3	DB	13.6	90.3	4.4	4.3	10.4	93.3	7.8	95.8	5.3	98.2	2.0	101.3	
			WB	12.8	73.5	4.4	1.7	9.7	74.7	7.1	75.7	4.6	76.6	1.5	77.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.8	34.3	36.1	43.1	50.3	58.0	64.4	67.3	66.6	58.8	51.5	42.1	36.0
	DBStd	13.57	7.39	7.50	6.93	7.31	6.66	6.96	6.04	5.70	5.81	6.19	6.89	6.92
	HDD50	1952	488	391	230	86	10	1	0	0	4	56	250	436
	HDD65	5511	952	809	680	442	233	95	46	49	200	418	687	900
	CDD50	2242	1	1	16	94	259	434	535	516	268	104	13	1
	CDD65	326	0	0	0	1	17	77	116	100	14	1	0	0
	CDH74	1885	0	0	0	9	94	486	716	537	41	2	0	0
	CDH80	396	0	0	0	1	10	106	164	113	2	0	0	0
	WSAvg	4.1	4.2	4.1	4.2	4.3	4.3	4.3	4.4	4.2	3.9	3.8	3.8	4.2
	PrecAvg	55.1	2.7	3.0	3.5	4.0	5.4	6.7	6.6	6.7	4.9	3.8	4.1	3.6
(15) Precipitation	PrecMax	70.8	5.9	8.3	7.0	7.4	12.6	12.4	12.2	11.5	7.9	8.2	9.4	7.6
	PrecMin	42.0	0.3	0.8	1.2	1.3	0.7	2.6	2.6	1.9	2.2	0.4	0.3	0.4
	PrecStd	6.1	1.6	1.6	1.5	1.7	2.6	2.4	2.6	2.4	1.5	1.7	2.0	1.7
	PrecStd	6.1	1.6	1.6	1.5	1.7	2.6	2.4	2.6	2.4	1.5	1.7	2.0	1.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	54.2	57.2	64.5	74.5	81.2	88.0	88.8	87.8	78.3	70.3	64.0	58.7	58.7
	0.4% MCWB	47.0	47.6	51.6	57.3	64.0	69.0	70.0	69.6	65.6	59.3	52.7	46.9	46.9
	2% DB	50.1	52.4	59.8	70.0	76.5	83.1	84.4	83.0	74.4	65.8	58.3	50.6	50.6
	2% MCWB	44.5	44.8	49.8	55.8	62.2	67.7	68.5	69.3	64.4	58.0	50.7	46.0	46.0
	5% DB	47.1	49.1	56.4	65.9	73.0	79.6	81.0	79.7	70.9	62.9	54.0	47.8	47.8
	5% MCWB	42.7	43.3	48.4	54.3	60.2	66.3	67.2	67.9	62.8	56.8	49.6	44.4	44.4
	10% DB	44.0	46.1	53.1	62.2	69.3	76.3	78.1	76.5	67.9	60.0	51.0	45.3	45.3
	10% MCWB	40.7	41.6	46.7	52.2	58.6	64.8	66.3	66.7	61.3	55.5	47.9	42.4	42.4
	0.4% WB	48.4	48.4	53.2	59.3	66.1	71.6	72.3	72.6	67.5	61.4	54.5	48.7	48.7
	0.4% MCDB	52.1	54.7	61.5	70.5	77.3	84.5	84.1	83.4	75.0	66.9	60.8	54.0	54.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	45.3	45.7	51.1	56.7	63.3	69.1	70.1	70.3	65.5	59.5	51.9	46.7	46.7
	2% MCDB	49.1	51.1	57.7	67.2	73.7	80.0	81.1	80.4	72.0	64.0	55.9	50.2	50.2
	5% WB	43.0	43.8	49.2	54.9	61.4	67.4	68.4	68.7	63.7	57.6	50.0	44.9	44.9
	5% MCDB	46.4	48.3	55.1	64.2	70.7	77.4	78.5	77.9	69.7	61.4	53.6	47.7	47.7
	10% WB	41.0	41.8	47.3	53.0	59.5	65.7	66.9	67.2	61.9	56.0	48.1	42.6	42.6
	10% MCDB	43.8	45.7	52.3	60.9	67.5	74.6	76.1	75.2	67.0	59.4	51.0	45.0	45.0
(35) Mean Daily Temperature Range	MDBR	8.0	10.0	12.7	14.8	14.7	14.9	15.0	14.7	13.1	10.6	8.7	7.7	7.7
	5% DB	11.2	14.7	18.3	21.1	20.5	20.4	20.1	19.7	17.4	14.6	13.6	11.4	11.4
	5% WB	7.6	9.1	10.3	10.4	9.2	9.0	8.7	9.0	8.1	7.9	7.4	7.4	7.4
	MCDBR	10.4	13.3	16.3	19.2	18.1	18.3	18.5	18.2	15.7	12.6	11.7	10.6	10.6
	MCWBR	7.7	8.6	9.6	9.9	8.8	8.6	8.8	8.8	8.1	7.7	7.5	7.2	7.2
	MCWBR	7.7	8.6	9.6	9.9	8.8	8.6	8.8	8.8	8.1	7.7	7.5	7.2	7.2
(40) Clear-Sky Solar Irradiance	taub	0.288	0.315	0.355	0.398	0.413	0.419	0.420	0.412	0.381	0.348	0.316	0.285	0.285
	taud	2.406	2.332	2.304	2.240	2.253	2.277	2.271	2.310	2.374	2.451	2.478	2.429	2.429
	Ebn at Noon	253	268	273	271	270	269	266	264	261	254	241	240	240
	Edh at Noon	22	29	35	41	42	41	41	38	32	26	20	19	19
(44) All-Sky Solar Radiation	RadAvg	374	646	1017	1366	1569	1788	1736	1523	1136	729	408	307	307
	RadStd	42	86	130	191	163	171	162	161	123	79	52	47	47

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+0.72	N/A	N/A	+2.09	N/A	N/A	N/A	-310	+211	+94
(47) Regional (107 neighbors)	N/A	N/A	N/A	N/A	+0.79	+0.70	N/A	N/A	+95	+23

Nomenclature: See separate page